Experiential Learning Project Assessment-I on

"TOWER OF HANOI VISUAL SIMULATOR"

under the course

Python ProgrammingLanguage(N-PCCCM304P)

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Problem statement

- Tower of Hanoi is a classic puzzle based on moving disks between three rods.
- The user selects the number of disks.
- All disks must be moved from the Source Rod to the Destination Rod.
- A larger disk cannot be placed on a smaller disk.
- The simulator displays each move step-by-step.
- It also calculates the minimum number of moves required.

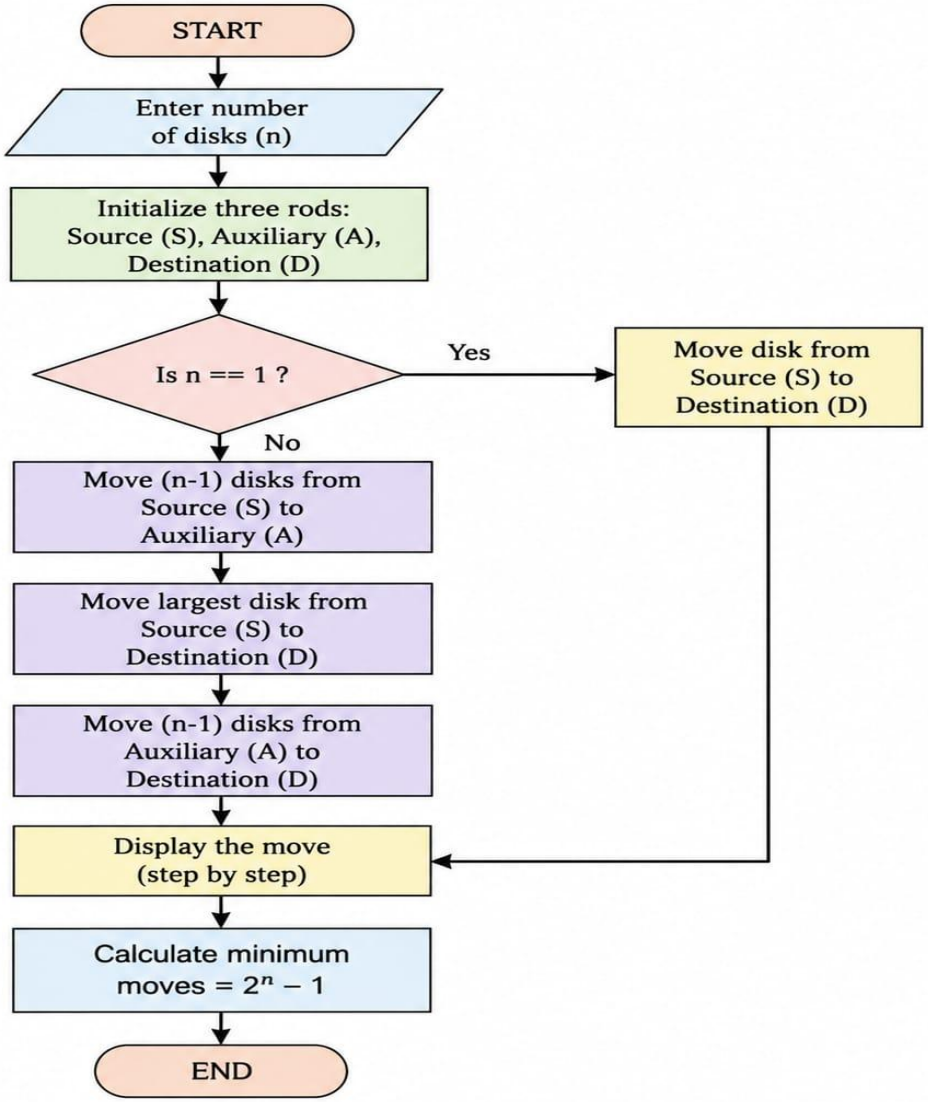
Objectives

- To understand the Tower of Hanoi problem.
- To implement the solution using Python.
- To use recursion for solving the puzzle.
- To visually display every disk movement.
- To calculate the minimum number of moves.
- To make the concept easy and interactive for users.

Algorithm

1. Start with n disks.
2. If there is only one disk, move it directly to the destination.
3. Move n-1 disks from Source to Auxiliary.
4. Move the largest disk from Source to Destination.
5. Move n-1 disks from Auxiliary to Destination.
6. Repeat until all disks are transferred.
7. Display every move sequentially.

Flowchart


START
Enter number of disks (n)
Initialize three rods: Source (S), Auxiliary (A), Destination (D)
Is n == 1 ?
Yes
Move disk from Source (S) to Destination (D)
No
Move (n-1) disks from Source (S) to Auxiliary (A)
Move largest disk from Source (S) to Destination (D)
Move (n-1) disks from Auxiliary (A) to Destination (D)
Display the move (step by step)
Calculate minimum moves = $2^n - 1$
END

Understanding of Python Logic

- Recursion: Solves the Tower of Hanoi problem.
- Functions: Separate the puzzle-solving and display operations.
- Loops: Used for displaying or processing moves.
- Lists: Can represent disks and rods.
- Conditional statements: Check the base case.
- GUI/Visualization: Can be used to show disk movement graphically.

GitHub Account, Repository Creation & Uploading Code

- Created a GitHub repository for the project.
- Uploaded the Python source code.
- Used meaningful commit messages.
- Updated the project regularly.
- Tested and corrected the program.
- Maintained individual contribution in the project.
- Github repository link

https://github.com/Tanvvviiii/Tower-of-Hanoi-Visual-Simulator/blob/main/tower_of_hanoi_visual_simulator_py

Conclusion

- The project successfully demonstrates the Tower of Hanoi puzzle.
- Python recursion makes the solution simple and efficient.
- The visual simulator helps users understand each movement.
- It calculates the minimum number of moves automatically.
- The project improves understanding of recursion, functions, lists and Python programming.

Thank You